ARCHIT AGRAWAL

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PROFESSIONAL EXPERIENCE

EdPlus, Arizona State University **06/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Designed S3 storage with compact fact tables and Lance payloads, queried through DuckDB for full-text and vector search.
- Parallelized data labeling across 64 AWS Step Functions lanes, with a failure circuit breaker to detect incomplete runs.
- Reached 100% join integrity on repeat runs with schema-constrained LLM output and per-row IDs; the prior pipeline silently mislabeled 1 in 8 rows.
- Accelerated publishing of a 51,280-respondent survey from 19 minutes to 109 seconds; an independent raw-data audit matched all evaluated figures.
- Built dataset intake for any table shape, from crosstabs and long tables to weighted surveys, including UTF-16 files and duplicate-header validation.
- Blocked double counting in cross-dataset joins by detecting runs that share the same people.
- Reduced AWS pipeline runtime from 500 to 85 seconds by replacing two managed-agent stages with direct tool calls.
- Added access-controlled row-level retrieval, AWS Bedrock Guardrails and per-stage spending limits to the analytics proof of concept.

MyStage Music Inc. **07/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Scaled ingestion to 70,000+ records daily from 650+ venues using Playwright, transactional locking and entity resolution.
- Made ingestion retries safe with deterministic task IDs and atomic create-if-absent writes, preventing duplicate jobs from overwriting in-flight work.
- Consolidated a 14-Cloud-Function backend into a checkpointed Python/LangGraph workflow on GCP Cloud Run, reducing production service dependencies 64%.
- Designed reusable scraping modules so new data sources can be added without changing the orchestration graph.
- Implemented checkpointed pause-and-resume so records missing context wait for additional information without failing the ingestion batch.

EdPlus, Arizona State University **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Implemented tenant isolation in retrieval services and configuration-based onboarding for additional colleges.
- Reduced transcript-analysis time from four hours to 15 minutes using a Neo4j knowledge graph and validated Cypher queries.
- Added hybrid keyword and vector retrieval to resolve exact course codes alongside semantic document search.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Built 20+ REST APIs in .NET 8 and an ETL pipeline ingesting 10,000+ researcher profiles daily into PostgreSQL.
- Made ETL reruns idempotent through natural-key upserts, preventing duplicate records after partial failures.
- Enforced uniqueness, not-null and range constraints in PostgreSQL to reject invalid records during ingestion.
- Cut API latency from 198 to 20 ms with Redis caching and SQL rewrites, verified under simulated peak load with Locust.
- Reduced p95 retrieval latency 60% at 95% recall using hybrid FAISS and Neo4j search.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Migrated a C#/.NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and resource footprint 35%.

- Reduced deployment time 70% and production incidents 40% through containerized CI/CD and SonarQube quality gates.
- Improved student-listing performance 30% with pagination and S3 asset delivery; reduced class-detail latency 10% through SQL query rewrites.

EAT.FIT **09/2021 - 12/2021**
Product Intern *Bengaluru, India*
- Built a WebSocket and Express notification service with backpressure-aware fan-out and duplicate-safe redelivery; reported CSAT rose 28% and support volume fell 35%.

PROJECTS

Image Recognition as a Service **01/2024 - 02/2024**
- Built an AWS EC2/SQS image-processing service with queue-based scaling, asynchronous results and idempotent workers for safe message retries.
- Capped worker capacity and used scale-in cooldowns to control cost and prevent instance churn during changing demand.

No-Code Pipeline Builder **2025**
- Built a React/ReactFlow workflow editor with typed nodes, undo/redo and autosave, backed by FastAPI graph validation.

TECHNICAL SKILLS

Languages & Storage: Python, SQL, TypeScript, PostgreSQL, DuckDB, S3, Lance, Redis
Pipelines: ETL, Step Functions, Lambda, GCP Cloud Run, FastAPI
Quality & Deployment: Data validation, idempotent processing, Docker, Kubernetes, CI/CD, pytest

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
M.S., Computer Science | GPA: 4.0 *Tempe, AZ*

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
B.Tech., Information and Communication Technology *Gandhinagar, India*